

EXPERIMENT 15

Implement probabilistic context-free grammar parsing for a sentence using python.

PROGRAM:

```
import nltk

from nltk import PCFG

from nltk.parse import ViterbiParser

grammar = PCFG.fromstring("""
S -> NP VP [1.0]
NP -> Det N [0.6] | 'John' [0.4]
VP -> V NP [1.0]
Det -> 'the' [1.0]
N -> 'dog' [0.5] | 'cat' [0.5]
V -> 'saw' [1.0]
""")

parser = ViterbiParser(grammar)

sentence = "John saw the dog".split()

for tree in parser.parse(sentence):

    print(tree)
```

OUTPUT:

```
)

parser = ViterbiParser(grammar)

sentence = "John saw the dog".split()

for tree in parser.parse(sentence):
    print(tree)

... (S (NP John) (VP (V saw) (NP (Det the) (N dog)))) (p=0.12)
```

